

Floor-Plan Reading and Rigid-Diaphragm Centre of Stiffness

Classical label supervision, two VLM readers, hand stiffness and an OpenSees check

Course	Computer (& AI) Methods for Structural Engineering (DISC 5203)
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Assigned plan	Floor Plan 1 (final digit 1)
Prepared	7 November 2026
Repository	Cumulative HW1 + HW2 + HW3 repository

18 elements | LLaVA 3/4 flags | CM (5.4605, 8.7915) m | CR (5.5488, 9.8946) m | OpenSees agreement 9.33e-10 m

Executive result

The required L09 order is preserved: fixed classical rules, one printed label per element, a retained but never-selected forest, then two VLMs restricted to frozen flag crops. LLaVA clears T4, T10 and T11; T3 remains unresolved and is excluded from the primary lateral model. The resulting eccentricity is 1.10660 m.

Rebuilding T3 at 309 mm and thickening T7 to 365 mm, without moving the mass layout or either centreline, reduces eccentricity to 0.000268 m (99.9758%). The same locked reader succeeds on NewPlan and fails safely on public MLStructFP966.

1. Fixed pipeline and audit boundary

Reader: grey<225; 11-px 45-degree opening; 17/21-px orthogonal openings; 4:1 shape rule; OCR twice at threshold 0.85. Pixels place; printed text sizes. VLMs receive local crops only and may not alter coordinates.

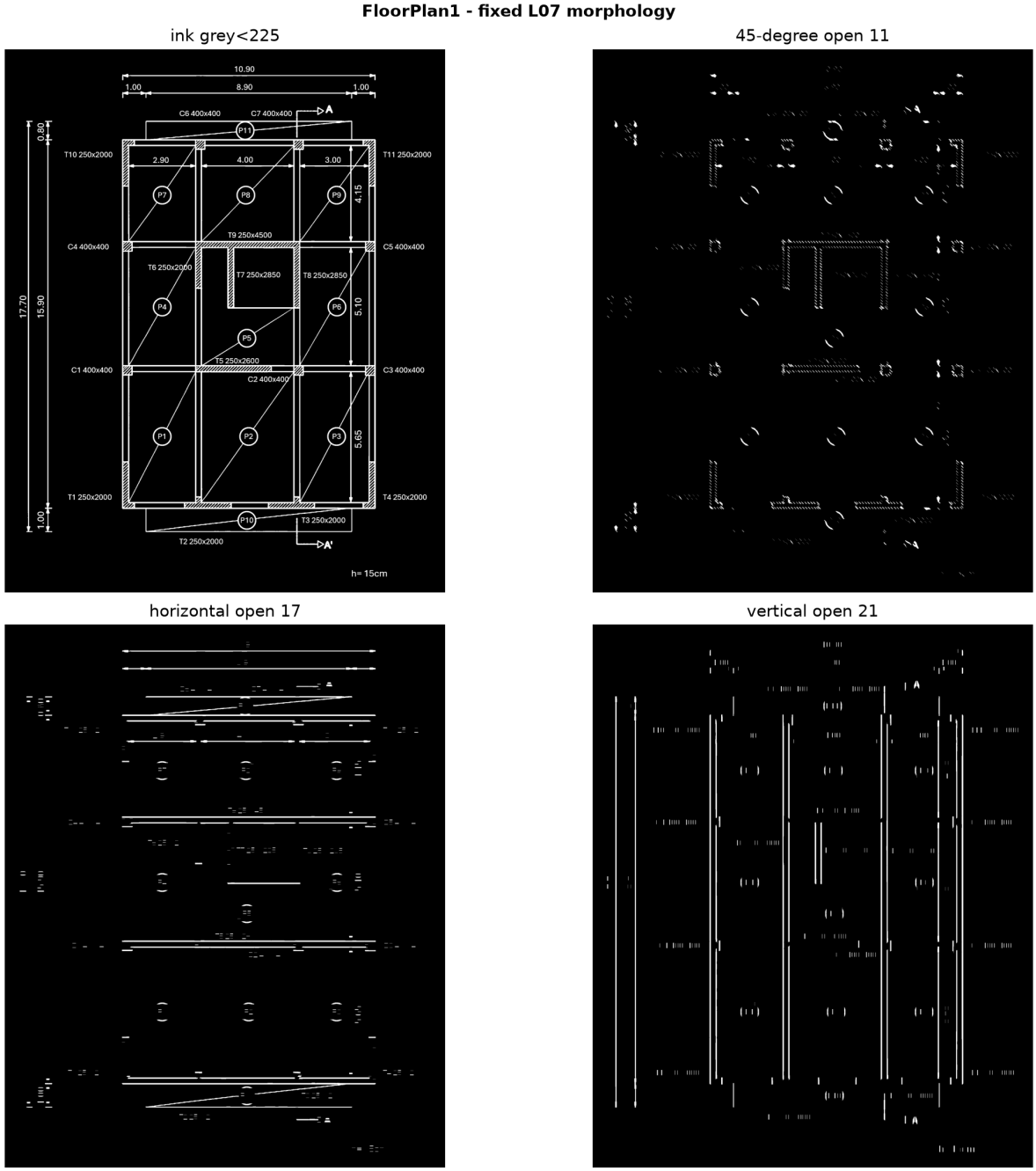


Figure 1. FloorPlan1 fixed threshold and morphology evidence.

The pass keeps 399 components and 68 elongated candidates. Both OCR passes recover all 18 names; label supervision closes 180.1285 m2 and 7880.622 kN on five storeys.

2. Label-supervised record and frozen flags

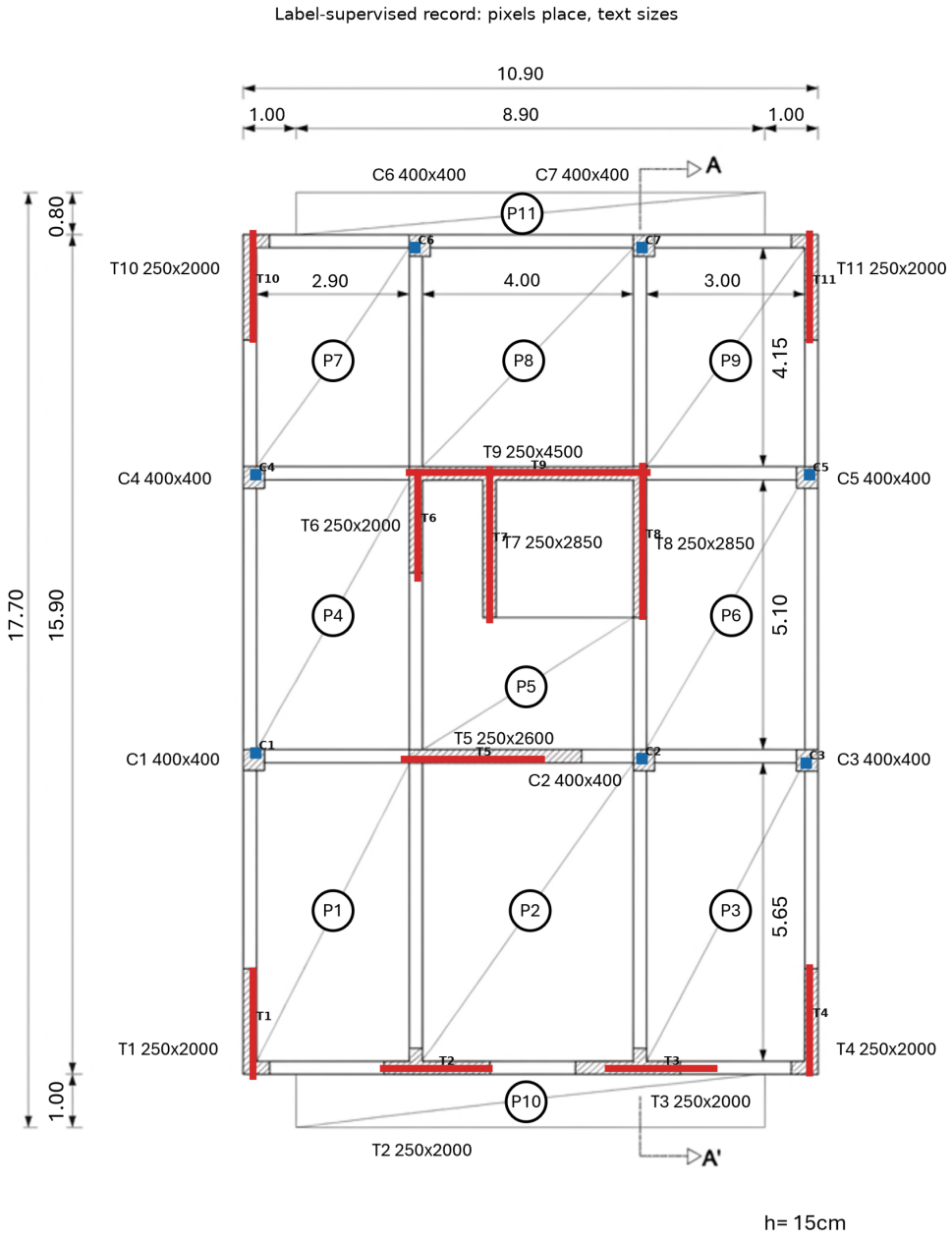


Figure 2. Immutable 7-column + 11-wall record; pixels place and text sizes.

Label	Classical flag	Expected size	Crop hash
T3	wall_in_pieces_along_axis+low_support_completion	250x2000	fd4c215dc5ec
T4	low_support_completion	250x2000	6b7ef25669c0
T10	low_support_completion	250x2000	5d3beb62b245
T11	low_support_completion	250x2000	05fb1b3ccb2c

The original JSON is never overwritten. Every crop, question, order, reference and confidence threshold is frozen before either model runs.

3. Kept-but-never-chosen random forests

Seven per-pixel features, 80 trees, depth 14, minimum leaf 3, balanced classes and seed 21345221 are identical across plans. Pseudo-labels come from the rules, so forest selection is prohibited.

Plan	OOB	Test F1	Rule IoU	Main evidence	Selected
FloorPlan1	0.982979	0.983485	0.584754	21/45 ink + 45deg	False
NewPlan	0.971208	0.970087	0.557338	21/45 ink + 45deg	False
MLStructFP966	0.746687	0.674299	0.221414	45-px ink	False

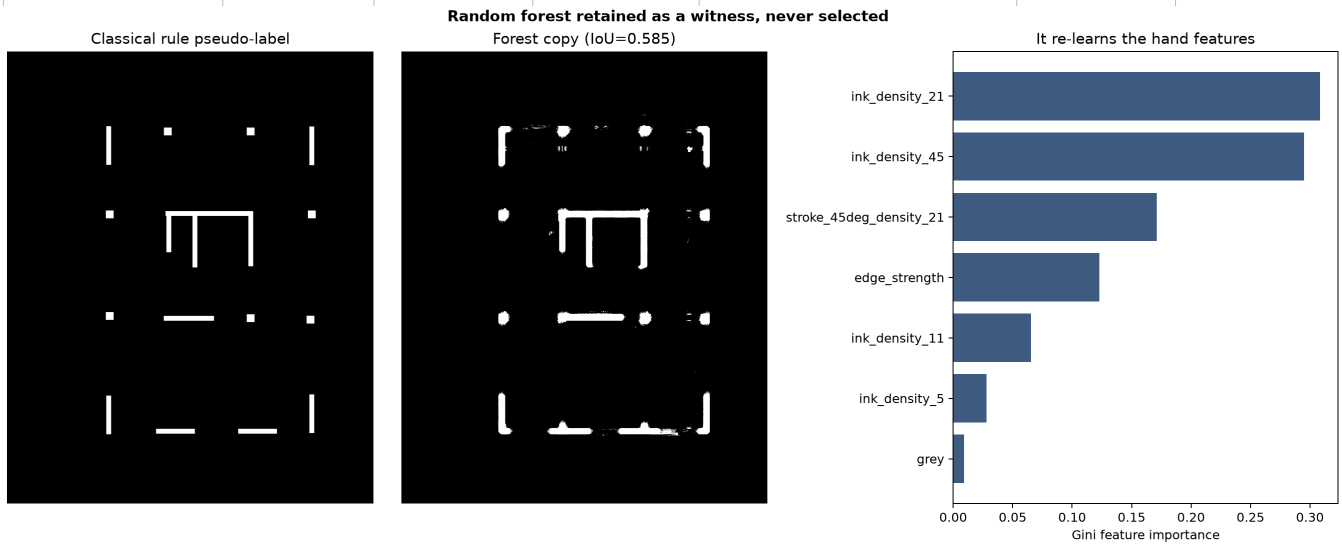


Figure 3. High held-out F1 does not create an independent correction route.

4. Two VLMs, one gate

Models: SmolVLM2-256M revision 067788... and LLaVA-OneVision-Qwen2-0.5B revision 74dd0b.... CPU float32, greedy decoding, seed 21345221. Required keys: label, size_mm, continuous, confidence, reason. A valid answer must exceed 0.85, agree with the independent reference and not contradict its own reason.

Plan	Model	N	Resolved	Broken	Unresolved	Conf	Err mm	Verdict
FloorPlan1	smolvlm2-256m	4	0	0	4	n/a	n/a	witness
FloorPlan1	llava-onevision-0.5b	4	3	0	1	0.9	0.0	llava-onevision-0.5b promoted
NewPlan	smolvlm2-256m	1	0	0	1	n/a	n/a	witness
NewPlan	llava-onevision-0.5b	1	1	0	0	0.9	0.0	llava-onevision-0.5b promoted
MLStructFP966	smolvlm2-256m	15	0	0	15	n/a	n/a	witness
MLStructFP966	llava-onevision-0.5b	15	0	0	15	n/a	n/a	witness

FloorPlan1: LLaVA clears T4/T10/T11, but T3 remains contradictory. Both actual VLM structural routes therefore sit 0.91416 m from the independent-reference CR. NewPlan: LLaVA clears its one OCR audit. MLStructFP966: both models resolve zero; the zero-resolved tie is rejected and no structural record is promoted.

Official model cards: https://huggingface.co/HuggingFaceTB/SmolVLM2-256M-Video-Instruct and https://huggingface.co/llava-hf/llava-onevision-qwen2-0.5b-ov-hf.

5. Hand CM and 3x3 stiffness

Wall stiffness includes bending and shear: k = 1/[H3/(3EI)+H/(G Av)], I=tL3/12 and Av=(5/6)tL. Signed slab regions give area 180.1285 m2, CM (5.460451, 8.791522) m, 160.665 t/storey and 803.325 t on five storeys.

Region	Area m2	cx	cy
main floor	173.3100	5.450	8.950
lower balcony	8.9000	5.450	0.500
upper balcony	7.1200	5.450	17.300
opening west	-3.3440	4.020	10.630
opening east	-5.8575	5.945	10.175

Primary hand matrices

```
K
[  3.6283e+06   0.0000e+00  -4.0023e+06 ]
[  0.0000e+00   4.0532e+06   3.5828e+05 ]
[ -4.0023e+06   3.5828e+05   1.1242e+08 ]

F=K^-1
[  2.8700e-07  -1.0000e-09   1.0000e-08 ]
[ -1.0000e-09   2.4700e-07  -1.0000e-09 ]
[  1.0000e-08  -1.0000e-09   9.0000e-09 ]
```

CR = (5.548845, 9.894589) m and e = 1.106603 m. Bending-only e = 2.012353 m; the shear term is material and retained.

6. Wall stiffness and 1000 kN push + turn

Wall	Dir	x	y	L	t	Active	k shear	Push	Turn	Total
T1	y	0.1911	1.9454	2.0	0.25	True	420875	0.0	-23.0	-23.0
T2	x	3.6537	1.0995	2.0	0.25	True	420875	116.0	37.8	153.8
T3	x	7.9113	1.0995	2.0	0.25	False	420875	0.0	0.0	0.0
T4	y	10.7242	2.0067	2.0	0.25	True	420875	0.0	22.3	22.3
T5	x	4.3493	6.9558	2.6	0.25	True	792157	218.3	23.8	242.1
T6	y	3.3097	11.3844	2.0	0.25	True	420875	0.0	-9.6	-9.6
T7	y	4.6627	11.0208	2.85	0.25	True	974408	0.0	-8.8	-8.8
T8	y	7.5597	11.082	2.85	0.25	True	974408	0.0	20.0	20.0
T9	x	5.3927	12.391	4.5	0.25	True	2415315	665.7	-61.6	604.1
T10	y	0.1911	15.9163	2.0	0.25	True	420875	0.0	-23.0	-23.0
T11	y	10.7242	15.901	2.0	0.25	True	420875	0.0	22.3	22.3

The +x load at CM equals a 1000 kN push through CR plus 1103.067 kN m torque. Forces close to 1000 kN in x, zero in cross-wall y and the applied torque.

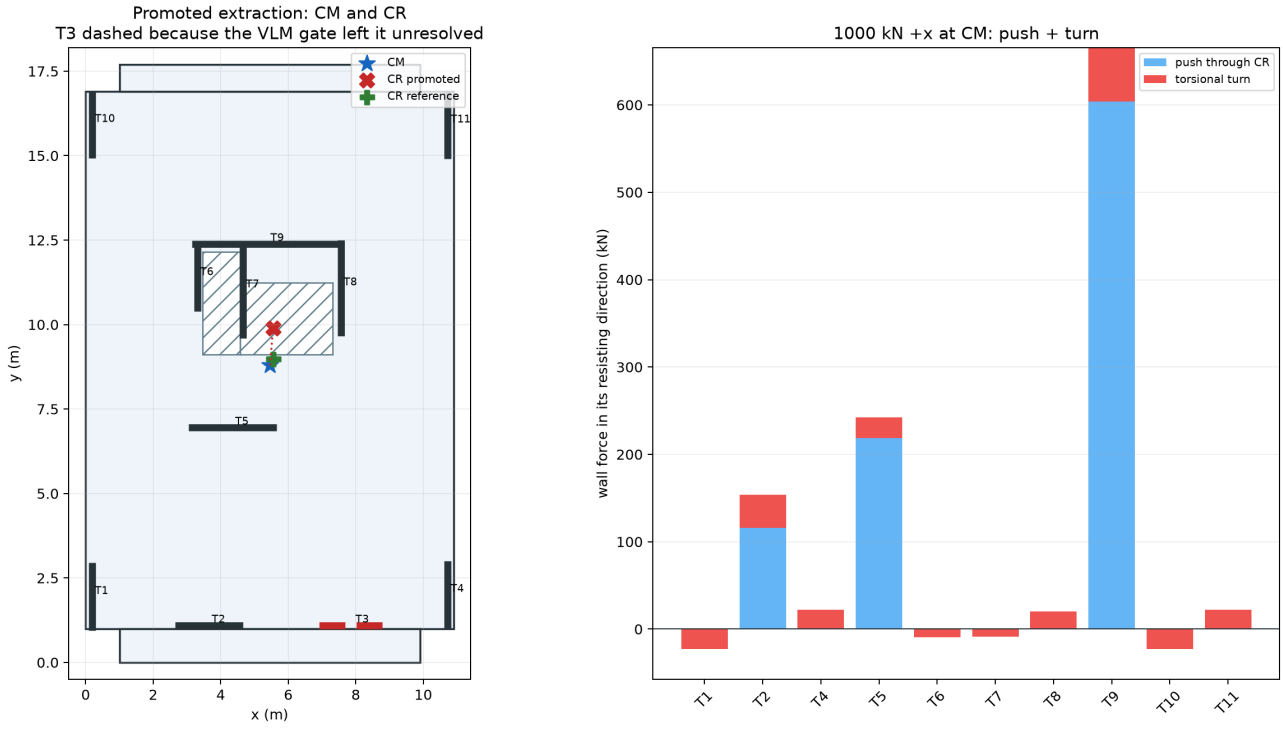


Figure 4. Main-plan CM/CR and wall-force decomposition.

7. OpenSees rigid-diaphragm check

One 3D ElasticTimoshenkoBeam per active wall; strong-axis I=tL3/12 and Av=(5/6)tL. Wall tops use rigidDiaphragm(3, master, *slaves) with Transformation constraints. Three unit loads at CM recover F and K.

```
OpenSees F
[  2.8688e-07  -9.0300e-10   1.0216e-08 ]
[ -9.0300e-10   2.4679e-07  -8.1900e-10 ]
[  1.0216e-08  -8.1900e-10   9.2610e-09 ]

OpenSees K
[  3.6283e+06  -6.4000e-11  -4.0023e+06 ]
[ -2.7000e-11   4.0532e+06   3.5828e+05 ]
[ -4.0023e+06   3.5828e+05   1.1242e+08 ]
```

Check	Result
CR difference from hand	9.330e-10 m
Maximum relative K difference	9.235e-09
Rotation for unit push through CR	-0.0 rad
1000 kN wall sum	999.999999 kN
Cross-wall sum	2.171e-08 kN
Push + turn max residual	0.0e+00 kN

Documentation: https://openseespydoc.readthedocs.io/en/latest/src/ElasticTimoshenkoBeam.html; https://openseespydoc.readthedocs.io/en/latest/src/rigidDiaphragm.html; https://openseespydoc.readthedocs.io/en/latest/src/TransformationMethod.html.

8. Make CM and CR coincide

Mass, openings, columns, grid and all wall centrelines remain fixed. T3 is rebuilt continuous at 309 mm; T7 is thickened from 250 to 365 mm.

Quantity	Before	After
CR x m	5.548845	5.460607
CR y m	9.894589	8.791740
Eccentricity m	1.106603	0.000268
1000 kN torque kN m	1103.067	0.218
Peak wall force kN	604.080	582.194
Ttheta/Tx	1.1068	1.0266

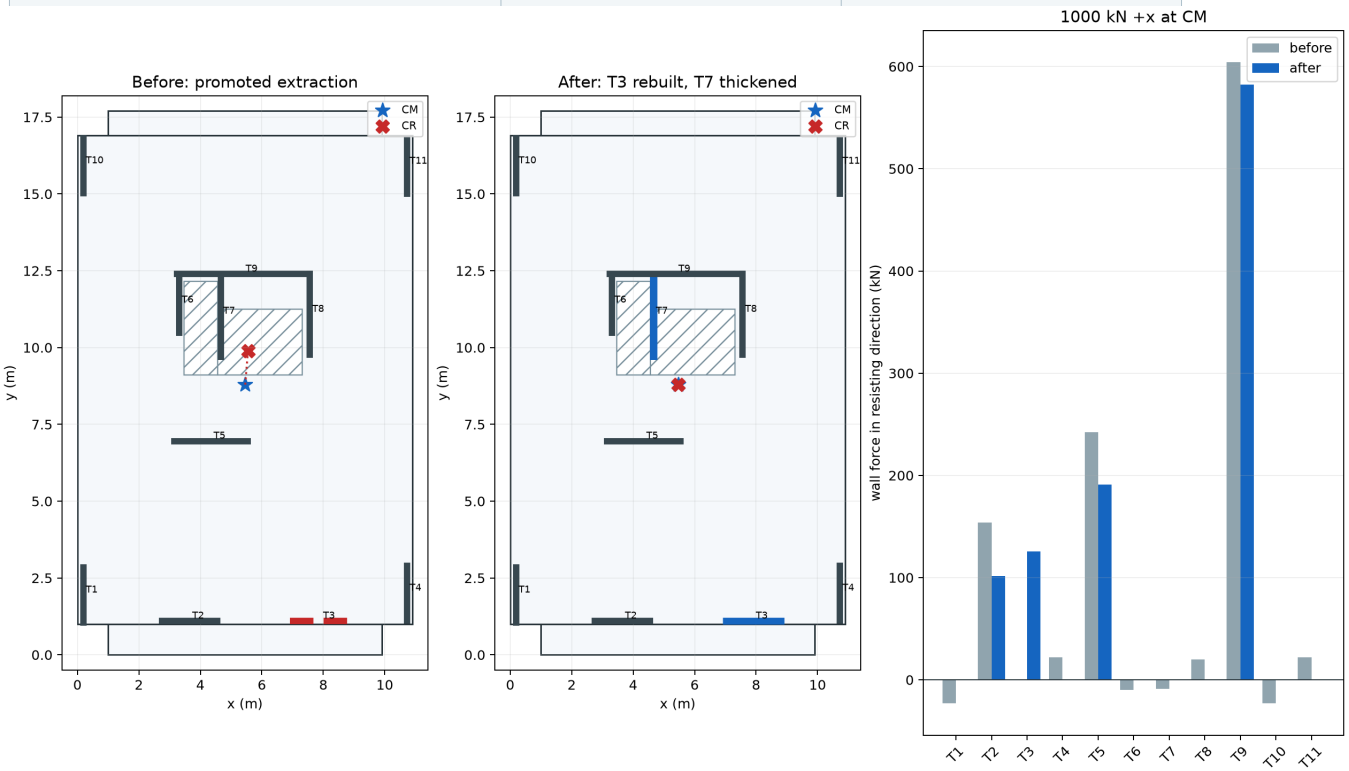


Figure 5. Same mass/grid; stiffness is retuned at two existing centrelines.

Eccentricity reduction = 99.9758%. Modified OpenSees CR difference = 1.55e-09 m.

9. Why coincidence matters

Before modification, e=1.10660 m creates 1103.07 kN m torque. The torsional radius is sqrt(29.758)=5.455 m; using 1 + e d/r_t^2, T2 is amplified to 1.3260 times its no-turn share. Peak absolute force falls 604.08 -> 582.19 kN.

Ttheta/Tx falls 1.1068 -> 1.0266. The pre-change first mode is 68.98% torsional; after decoupling it is nearly pure torsion rather than mixed translation-torsion. Coincidence removes coupling but need not reorder modes.

CM=CR removes torque for one uniform floor under a uniform lateral action. It does not remove torsion from height-varying layouts, nonuniform ground motion, cracking, uncertainty or higher modes. Extraction matters: the unresolved T3 shifts the VLM route 0.914 m from the independent-reference CR.

Key checks

Check	Before	After
Unit push through CR rotation	0	0
Push through CM rotation	nonzero	near zero after e reduction
Push + turn superposition	exact to saved precision	exact to saved precision
Mass layout changed	no	no

10. Course holdout: NewPlan

Unchanged parameters recover 38/38 names: 30 columns, 8 walls. T1 is reconstructed from split OCR and is the only flag. LLaVA resolves 1/1 at 0.9; Smol resolves 0/1. Area closes 828.0 m2 and five-storey axial load closes 36,225 kN.

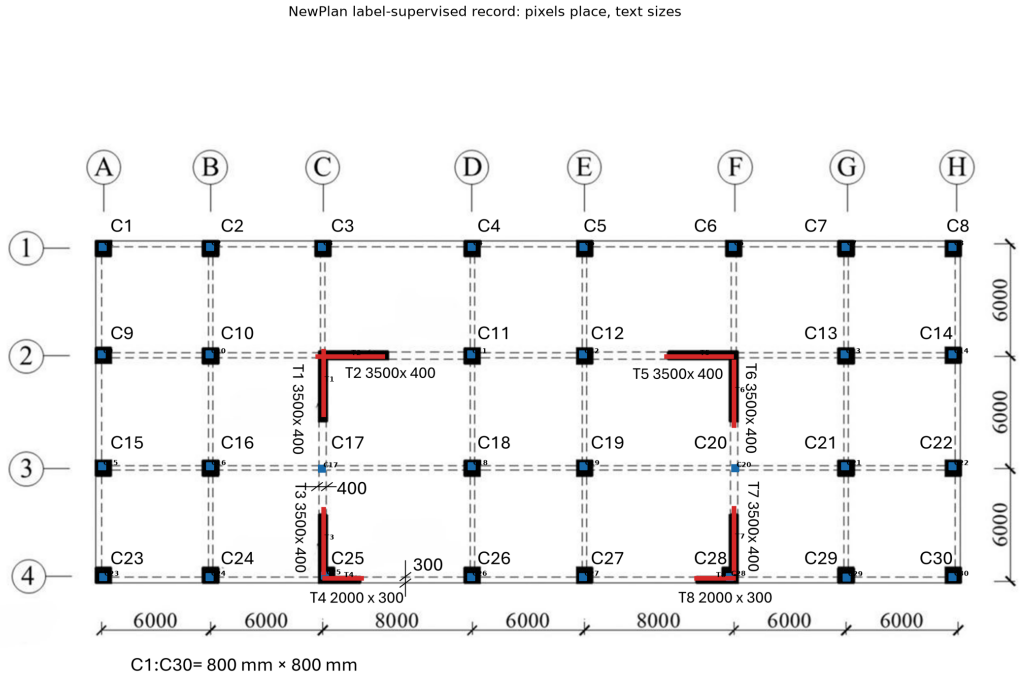


Figure 6. NewPlan 38-element label-supervised record.

CM=(23.0000, 9.0000) m; CR=(23.00745, 9.94591) m; e=0.94594 m.

11. NewPlan structural result

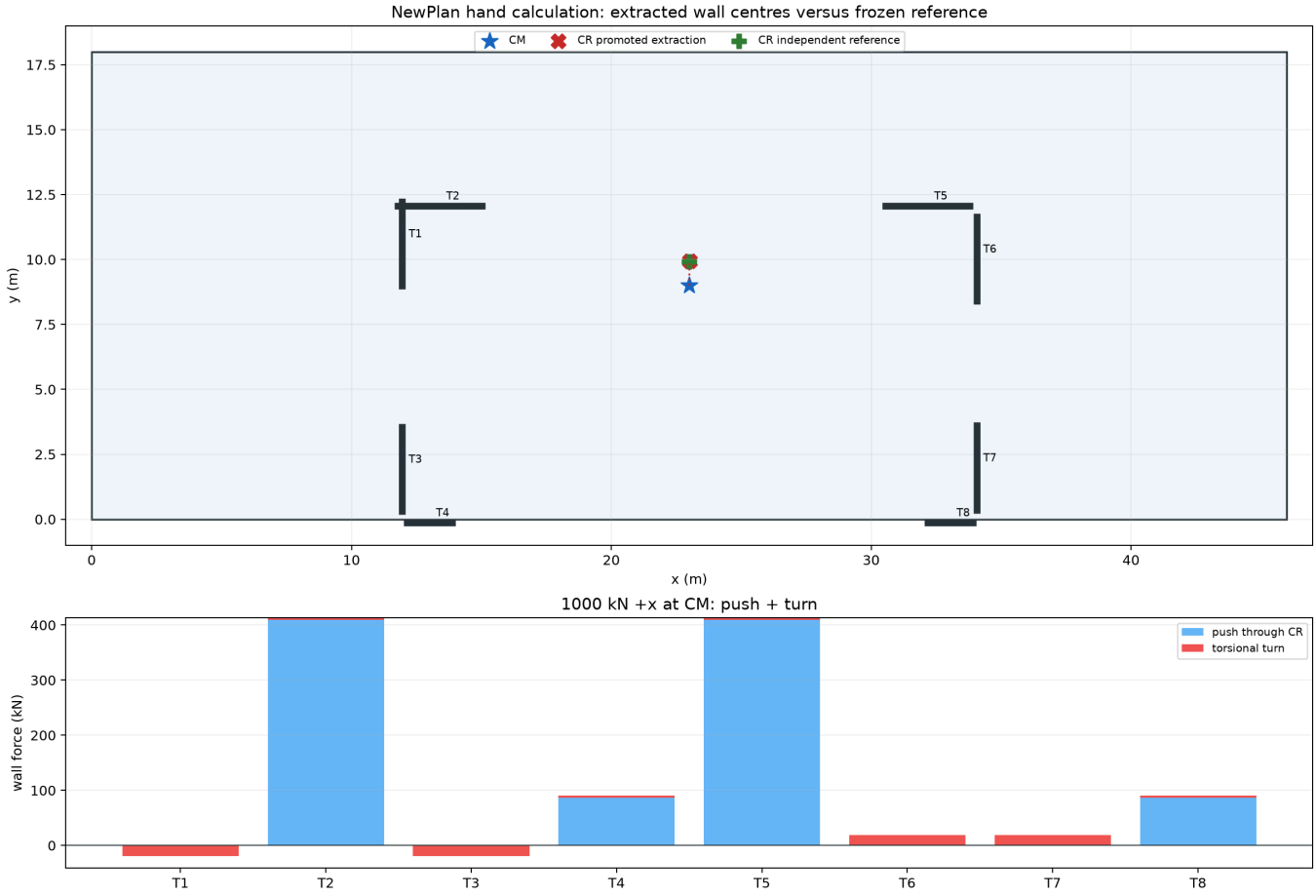


Figure 7. Extracted and frozen-reference CR differ by 28.8 mm.

T1's classical continuity was already true; its flag concerned split OCR. Both VLM structural states therefore retain eight walls. The VLM benefit is independent audit certainty, not coordinate or stiffness change.

12. Public holdout: MLStructFP Floor 966

Source: public MLStructFP test Floor 966 (MIT), 9600x6784 transparent blue-line CAD, irregular slab, 20 wall groups/28 rectangles, one 8.5-degree wall and zero printed C/T size strings. https://github.com/MLSTRUCT/MLStructFP; https://doi.org/10.1016/j.autcon.2023.105132.

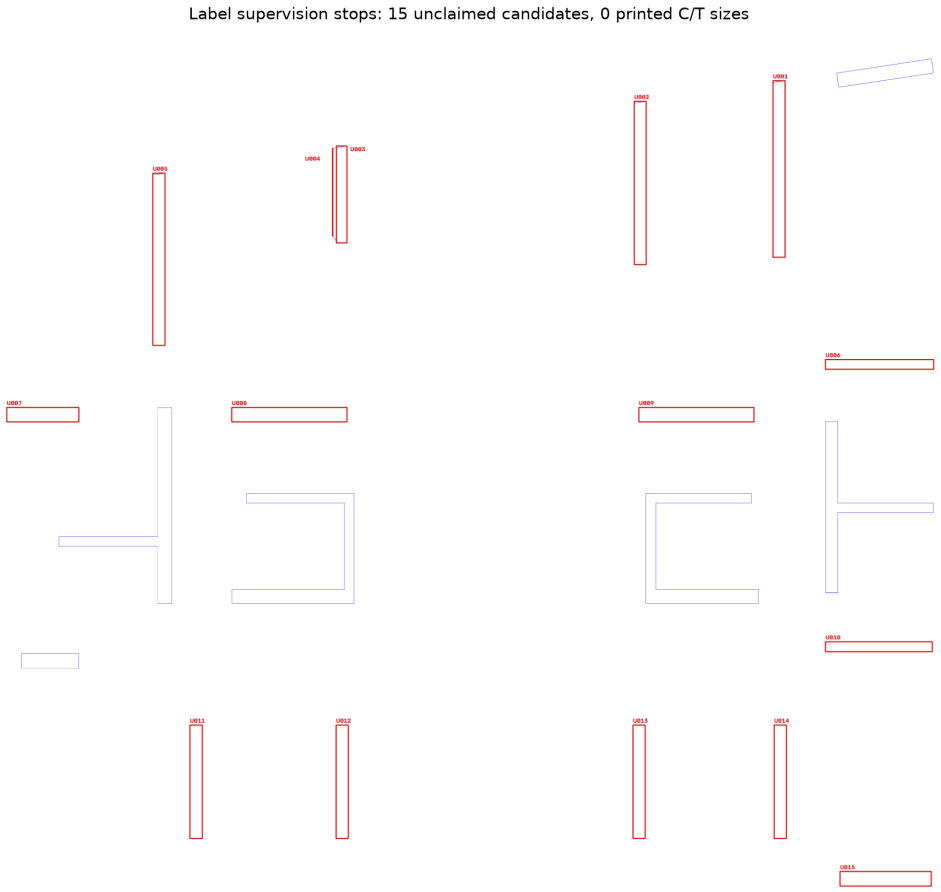


Figure 8. Fixed morphology: 15 unclaimed candidates, but zero auditable elements.

Metric	Result
Connected components	20
Elongated candidates	15
Accepted OCR labels	0
Claimed elements	0
Unique public rectangles hit	14/28 = 50%
VLM result	both 0/15; no promoted reader

13. Public-plan forest and safe failure

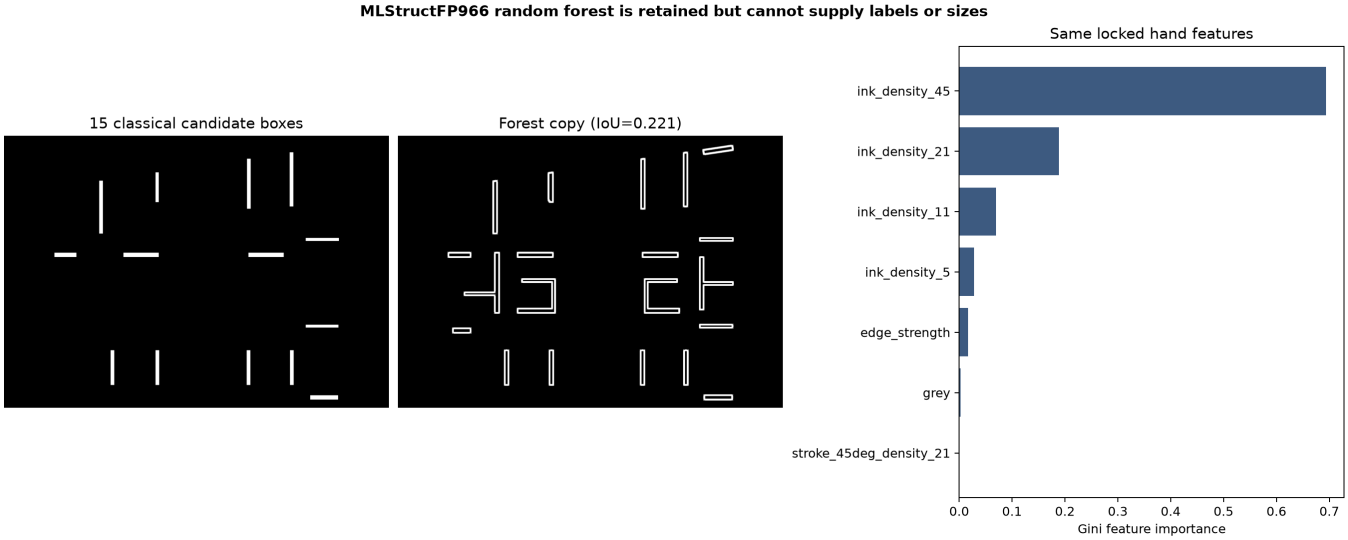


Figure 9. Forest F1 0.6743 and full-image IoU 0.2214; it copies rejected composites.

The forest has no printed IDs/sizes and is never selected. Both VLMs resolve 0/15. NOT_VISIBLE is not a wall size; therefore the accepted pipeline CM, CR and e are unavailable. The result is a safe failure rather than public-annotation leakage.

Reference-only benchmark

Using the frozen public slab and all 28 frozen rectangles only for comparison: CM=(14.220398, -9.818857) m; CR=(13.266821, -9.988242) m; e=0.968504 m. These are not pipeline outputs.

14. Public reference benchmark

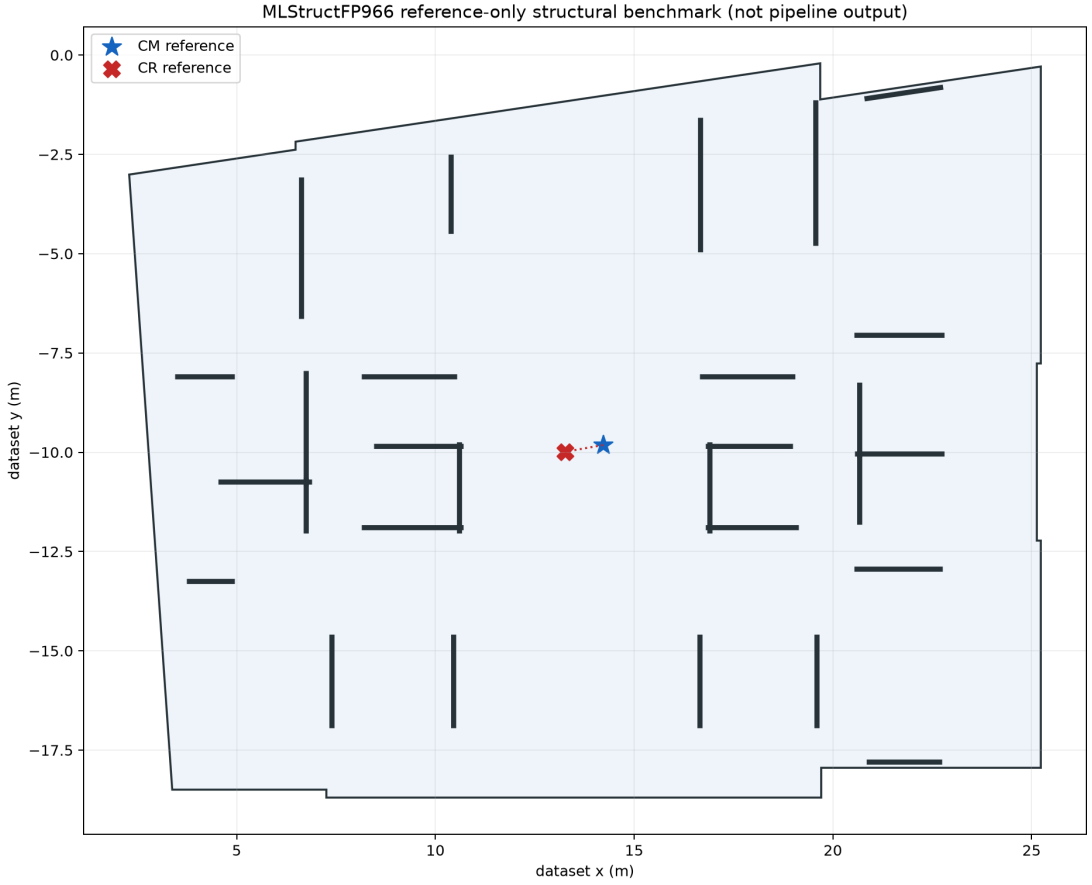


Figure 10. Reference-only CM/CR; title prevents confusion with extraction output.

The arbitrary-angle vector is [cos(a), sin(a), -cos(a)(y-ym)+sin(a)(x-xm)]. The transformed CR matrix has zero translation-rotation coupling to saved precision; a 1000 kN +x reference load closes force and moment equilibrium.

15. Cross-plan verdict

Plan	Elements	Flags	Best VLM	Pipeline CM	Pipeline CR	e	Verdict
FloorPlan1	18	4	LLaVA 3/4	(5.4605,8.7915)	(5.5488,9.8946)	1.1066	accepted
NewPlan	38	1	LLaVA 1/1	(23.0000,9.0000)	(23.0075,9.9459)	0.9459	accepted
MLStructFP966	0	15	none 0/15	unavailable	unavailable	unavailable	safe failure

What generalized: frozen morphology on isolated bars, deterministic forests, crop hashes, a non-regressing VLM gate and a closing structural calculation when numeric geometry exists. What did not: the one-label/one-element premise on no-text CAD, 4:1 separation of composite walls, and small-model protocol following.

Reproducibility

Scripts regenerate every deterministic output; real VLM outputs are committed and routine checks use --score-only. OpenSees runs in the isolated Python 3.12 target. The full command sequence and deliverable index are in README-HW3.md and the Markdown report.

Limitations

Raster OCR/morphology are style-sensitive. Structural models are linear, elastic, uncracked and single-storey; diaphragm flexibility, nonlinear cracking, soil, vertical irregularity and nonuniform excitation are excluded. Public annotations are defensible references, not truth.

Deliverables

Markdown + PDF report; immutable and promoted records; both VLM outputs per plan; three retained forests; hand/OpenSees JSON and wall-force CSVs; figures 01-19; exact requirements; and the complete material iteration log.